

1. Turn on 14/15 cameras by plugging in ethernet switch power to the extension cable
2. Turn on 1/15 cameras by turning on a separate power supply on the desk.
  - a. NO TRACKING
  - b. 0.45 A, 12 V
  - c. Turn on output
  - d. Verify this ethernet cable is plugged into the back of the computer using the white ethernet-USB dongle
3. Wait 1 hour to heat up cameras
4. Verify the even/odd kinematic mounts are correct on metrology stand (even means cone is by the computer and the cone on the aluminum HC)
5. Verify the diffuser for the top ball is on correctly
  - a. This is screwed on for a un-cabled station
  - b. This is hot-melted for a cabled station
6. Insert station and copper HC using spatula
7. Complete 3rd point
8. Wait at minimum 10 minutes for station to settle once landed
9. Attach the two station fiducials+diffusers on the arms (use gloves)
  - a. If this is a cabled station, make sure no cables are jutting out from the check valve holders into the images
10. Attach the top station fiducial (use the yellow stepper+gloves)- this fiducial has a black washer, do not drop the ball or the washer (the washer is loose). Verify this washer is flush with the bottom of the DCDC converter block
  - a. If this is a cabled station, make sure the black braided cable is not in the way of the ball, if it is, temporarily tape it down, make sure no other cables are in the way of the image
11. Insert the arduino black power cable into arduino on the copper HC
12. Power on copper HC power supply
  - a. TRACKING ON
  - b. A limit to 0.16 A
  - c. V limit to 22.8 V for triplet LEDs, V limit to 23.0 V for station LEDs
  - d. Do not turn on yet
13. Connect red/black power cables from the arduino to the copper HC power supply
  - a. Red cable goes to red, black cable goes to blue (all the way on the left end)
  - b. Still do not turn on power supply current
14. Make the appropriate directories: e.g. Production16\_09\_18\_CalibArm\_UltemBase , Production16\_09\_18\_CalibArm\_UltemBaseBig or Production14\_09\_18\_CalibArm\_UltemBase\_Cabled
15. Open the README file
16. Copy the relevant template command either for triplet LEDs or for station LEDs
  - a. Open up a blank text file
  - b. Paste the relevant template command
  - c. edit the directory (including the slash), edit the pythonw to be python3 (or just edit this once in the README)
  - d. copy the command

- e. paste the command in the command line, do not enter the command
- 17. Verify that all the cameras (14/15 using the switch and 1 using the power supply) are connected to power and have been sitting for 1 hour
- 18. Verify the arduino is powered by the computer and the LED circuit is connected to the power supply
- 19. Verify the station diffusers and station fiducials are in place
- 20. Send current to on copper HC (turn output on)
- 21. Press enter with relevant command, once command is finished, turn output off sent to copper HC
- 22. Visually verify the images look ok in folder and code did not crash
- 23. Upload images to onedrive
- 24. Remove station fiducials and diffusers (all 3) with gloves
- 25. Disconnect copper HC power supply cables (red/black) from power supply, hook up to copper HC using the tape
- 26. Remove black arduino power cable from Arduino on copper HC
- 27. Turn off all cameras (turn off voltage of power supply, then turn off), remove ethernet switch power from power strip